

1. AIM

To design and verify the schematic and layout of a Dynamic CMOS 2 input nand using 180 nm technology, and compare its pre-layout and post-layout simulation results

2. TOOL

Cadence Virtuoso

3. THEORY

A 2-input Dynamic CMOS NAND gate operates in two distinct phases: precharge and evaluation, controlled by a clock signal.

During the precharge phase, when the clock is LOW, the precharge PMOS transistor is ON and charges the output node to V_{dd} . The pull-down network (PDN) is disabled during this phase, so the output is precharged to logic HIGH.

During the evaluation phase, when the clock becomes HIGH, the precharge transistor is turned OFF and the PDN is enabled. The PDN consists of two NMOS transistors connected in series and controlled by inputs A and B . The output discharges to LOW only when both A and B are HIGH. For any other input combination, the output retains its precharged HIGH state.

The logic function of the 2-input Dynamic CMOS NAND gate is $Y = A \cdot B$. Since the output is already precharged to HIGH before evaluation, no LOW-to-HIGH transition occurs during evaluation. Therefore, $t_{pLH} = 0$ for the evaluation operation, while the relevant delay is t_{pHL} during discharge through the PDN.

The schematic of the 2-input Dynamic CMOS NAND gate is designed and simulated using 180 nm CMOS technology in Cadence Virtuoso. The corresponding physical layout is designed and verified using design-rule checking (DRC) and layout-versus-schematic (LVS) verification.

Pre-layout and post-layout simulations are then performed and compared to analyze the effect of layout parasitic on propagation delay and overall circuit performance.

4. TABLES

Table 4.1: Design Parameters

Parameter	Value	Unit
Technology	180	nm
V_{dd}	1.8	V
PMOS Width	400	nm
NMOS Width	400	nm

5. CALCULATION

$$N_{MH} = V_{OH} - V_{IH} = 1.8 - 0.836654 = 0.963346V \approx 0.9633V$$

$$N_{ML} = V_{IL} - V_{OL} = 0.555495 - 0 = 0.555495V$$

$$N_{MH(post)} = V_{OH} - V_{IH} = 1.8 - 0.8369212 = 0.9630788V \approx 0.9631V$$

$$N_{ML(post)} = V_{IL} - V_{OL} = 0.5554912 - 0 = 0.5554912V$$

$$t_{HL(pre)} = (37.0219 - 37.0075) * 1000 = 14.4ps$$

$$t_{HL(post)} = (37.0302 - 37.0075) * 1000 = 22.7ps$$

$$t_{LH(pre)} = 0ps$$

$$t_{LH(post)} = 0ps$$

$$t_{p(pre)} = 37.0219ps$$

$$t_{p(post)} = 37.0302ps$$

$$t_{r(pre)} = (20.0648 - 20.0185) * 1000 = 46.3ps$$

$$t_{r(post)} = (20.1034 - 20.0234) * 1000 = 80.0ps$$

$$t_{f(pre)} = (37.0376 - 37.0137) * 1000 = 23.9ps$$

$$t_{f(post)} = (37.056 - 37.0154) * 1000 = 40.6ps$$

$$Error_{V_{th}} = (0.0004/715.711) * 100 = 0.0001$$

$$Error_{V_{IH}} = (0.2672/836.654) * 100 = 0.0319$$

$$Error_{V_{IL}} = (0.0038/555.495) * 100 = 0.0007$$

$$Error_{N_{MH}} = (0.0002/0.9633) * 100 = 0.0208$$

$$Error_{N_{ML}} = (0.0000038/0.555495) * 100 = 0.0007$$

$$Error_{t_p} = (0.0083/37.0219) * 100 = 0.0224$$

CIRCUIT DIAGRAMS & WAVEFORMS

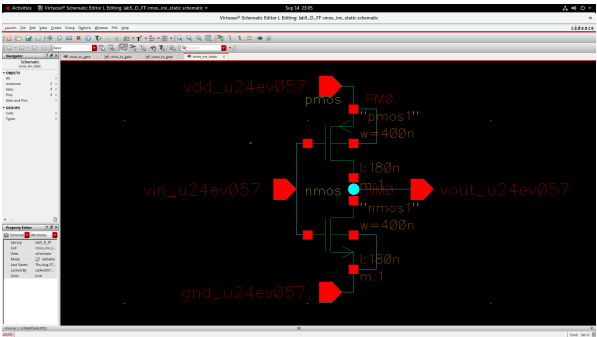


Figure .1: Inv Schematic

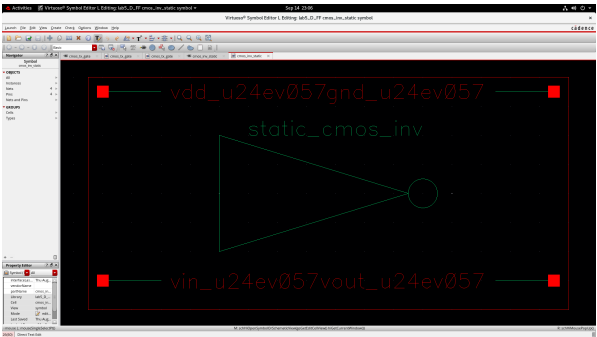


Figure .2: Inv Symbol

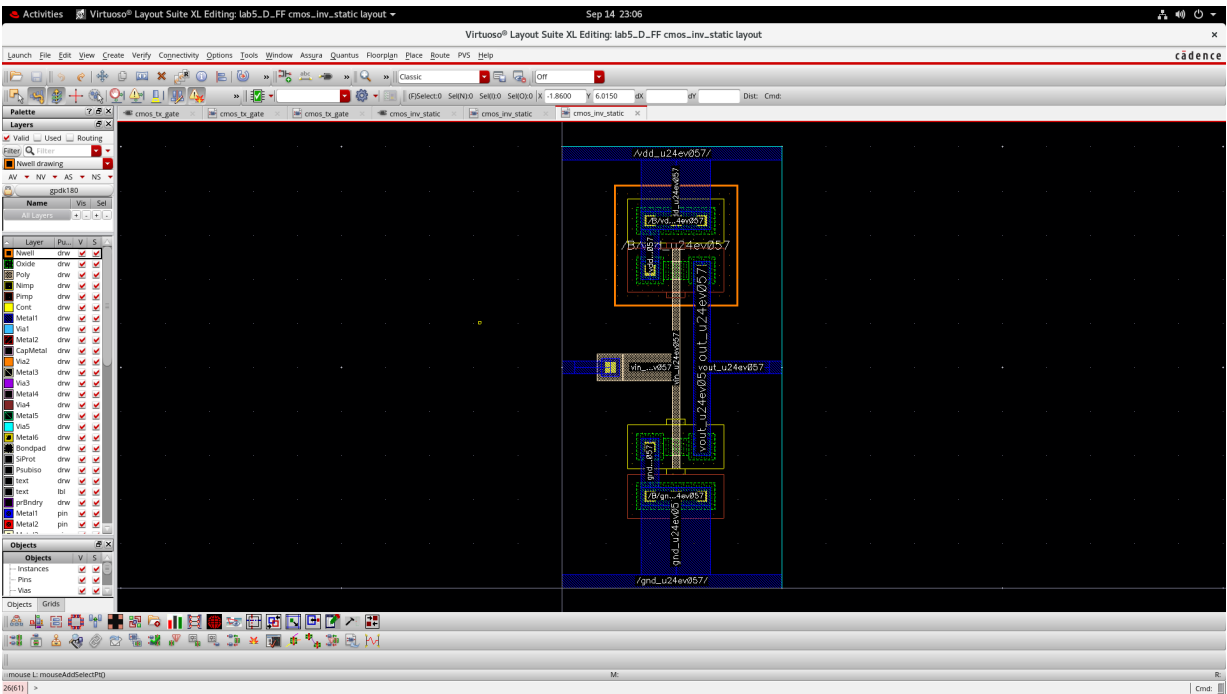


Figure .3: Inv Layout

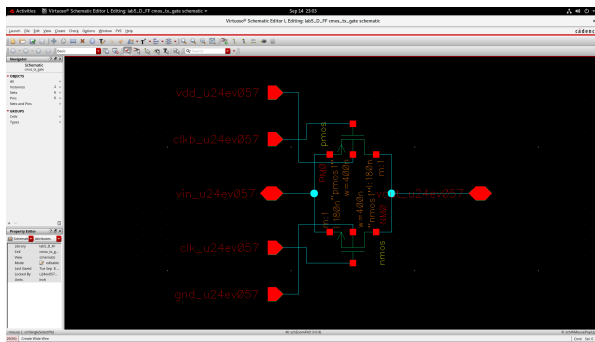


Figure .4: Tx Gate Schematic

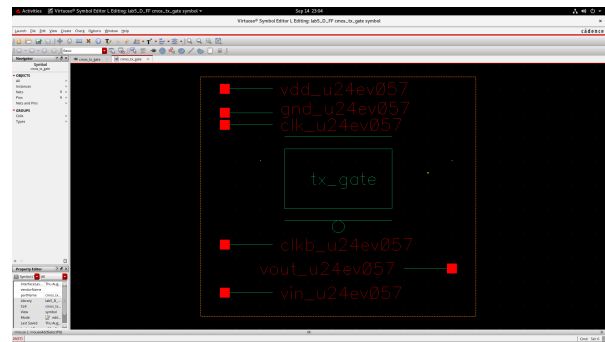


Figure .5: Tx Gate Symbol

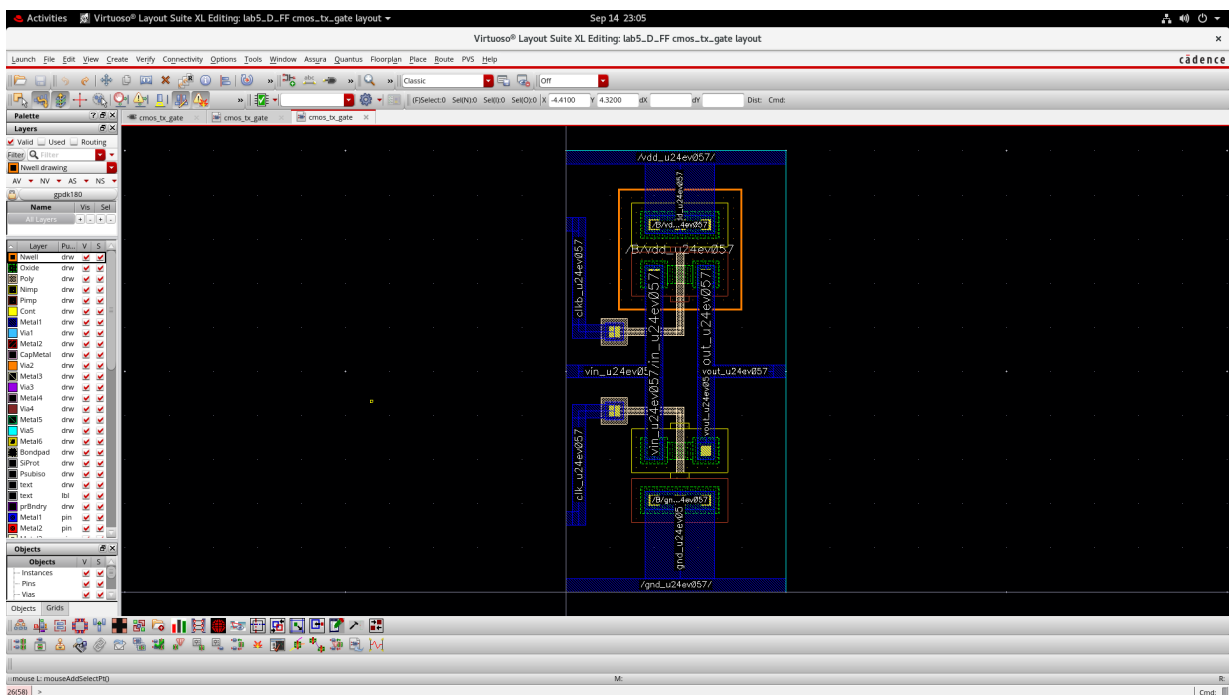


Figure .6: Tx Gate Layout

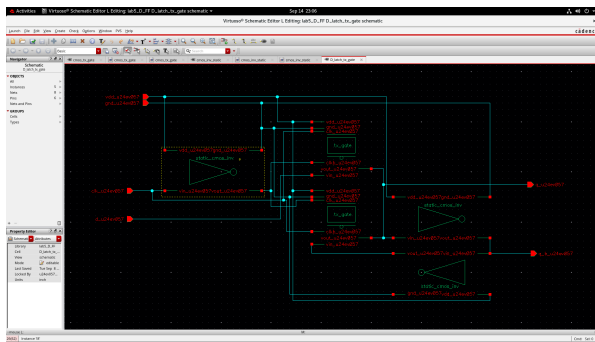


Figure .7: Tx Dlatch Schematic

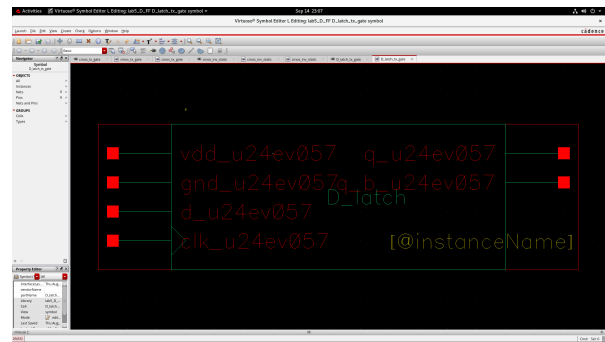


Figure .8: Tx Dlatch Symbol

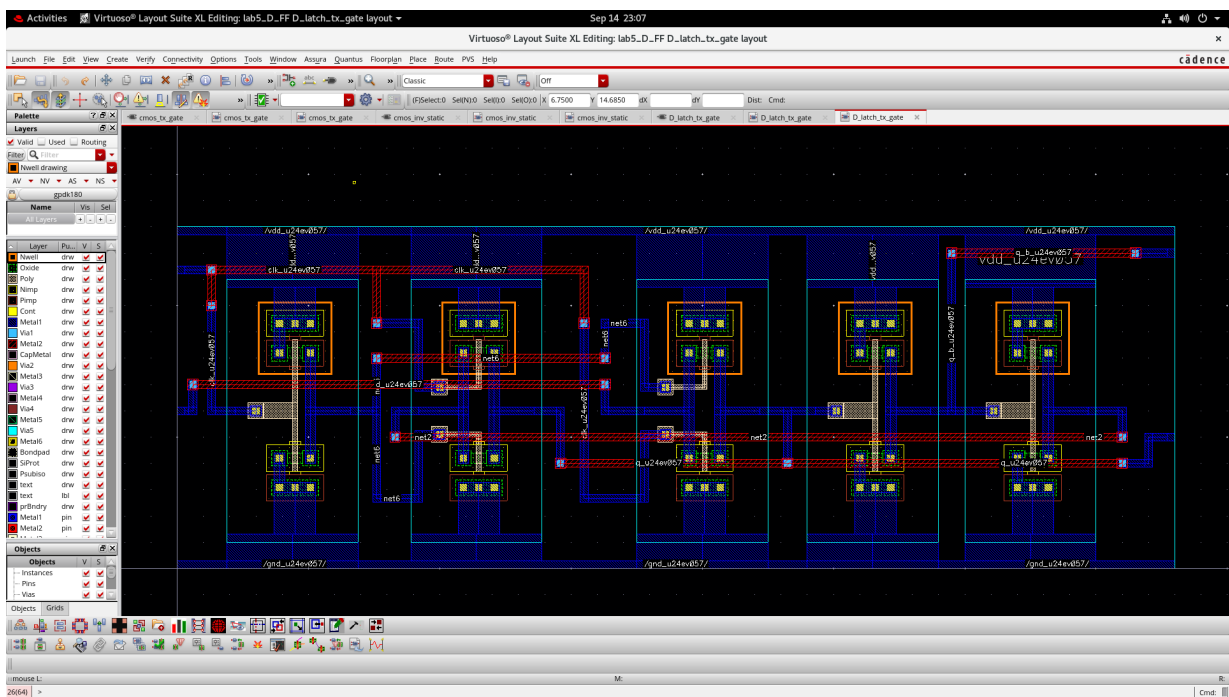


Figure .9: Tx Dlatch Layout

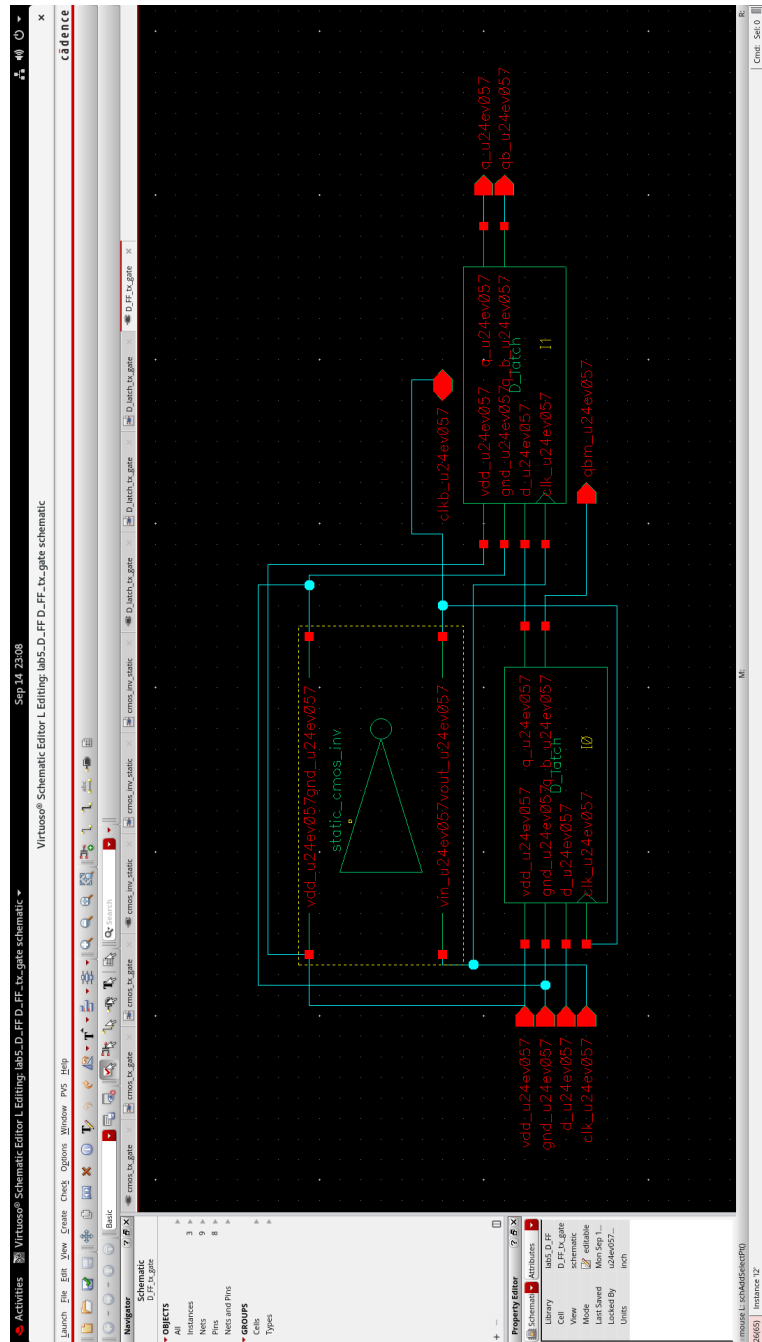


Figure .10: Tx Dff Schematic

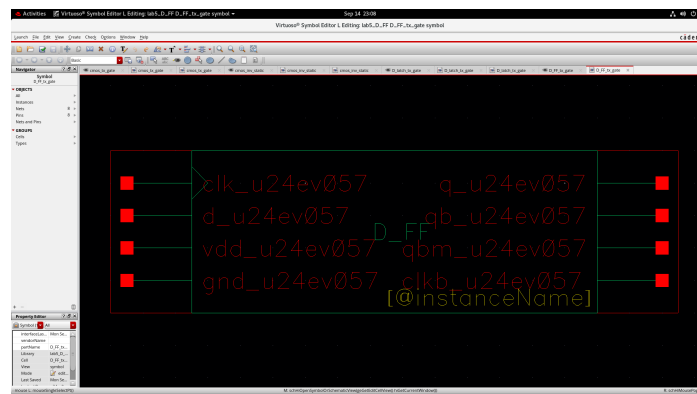


Figure .11: Tx Dff Symbol

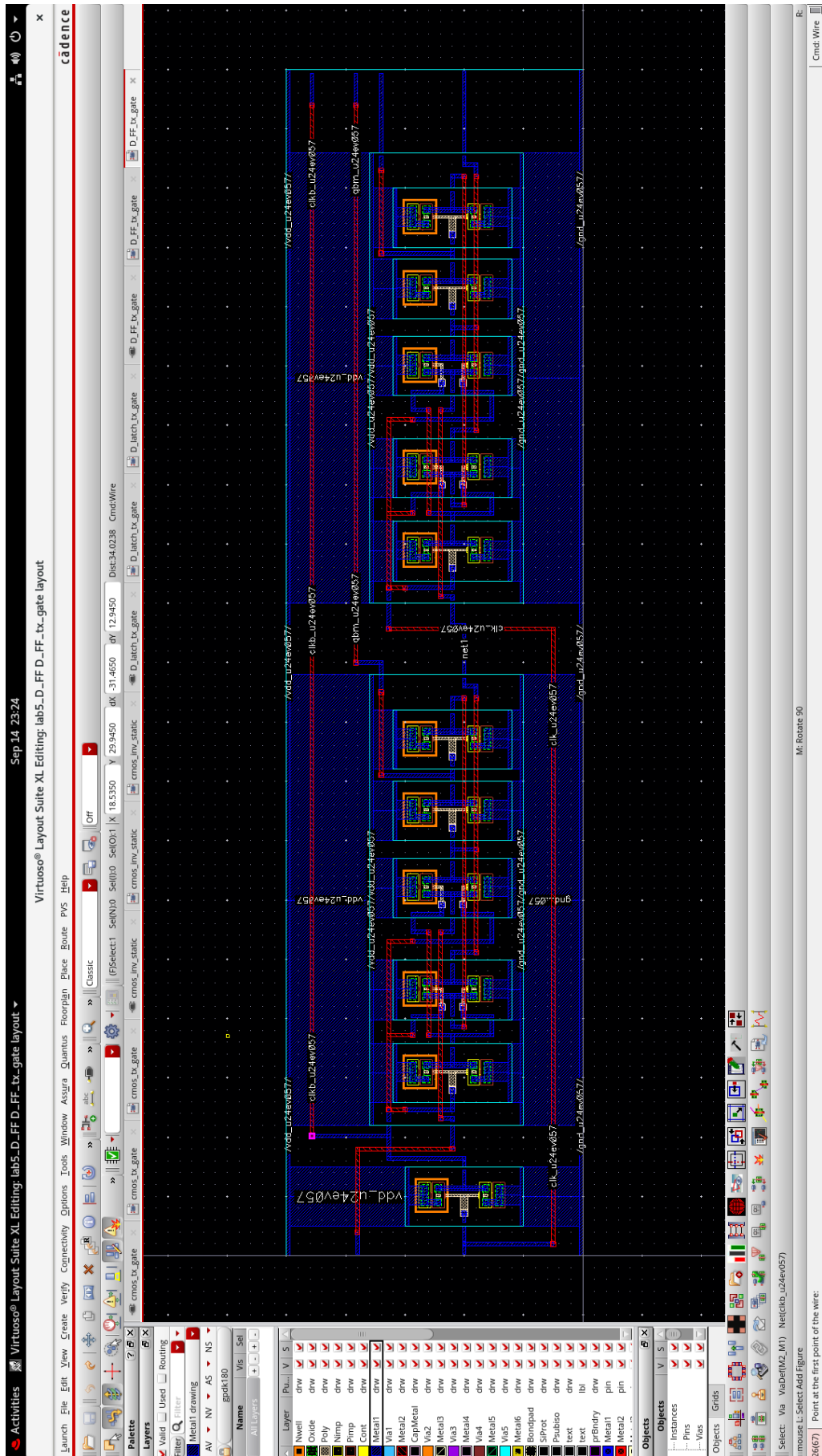


Figure .12: Tx Dff Layout

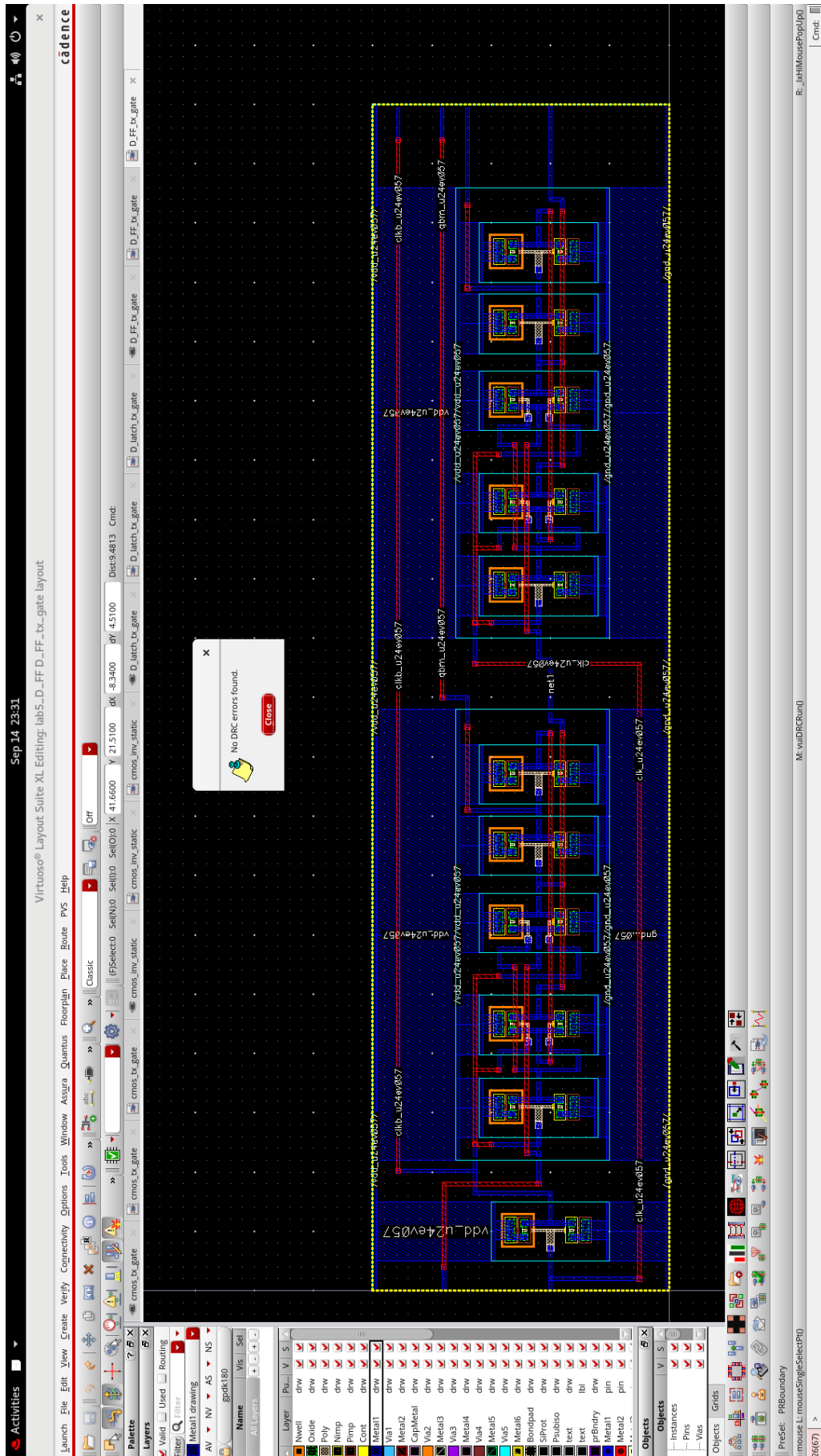


Figure .13: Tx Dff Drc

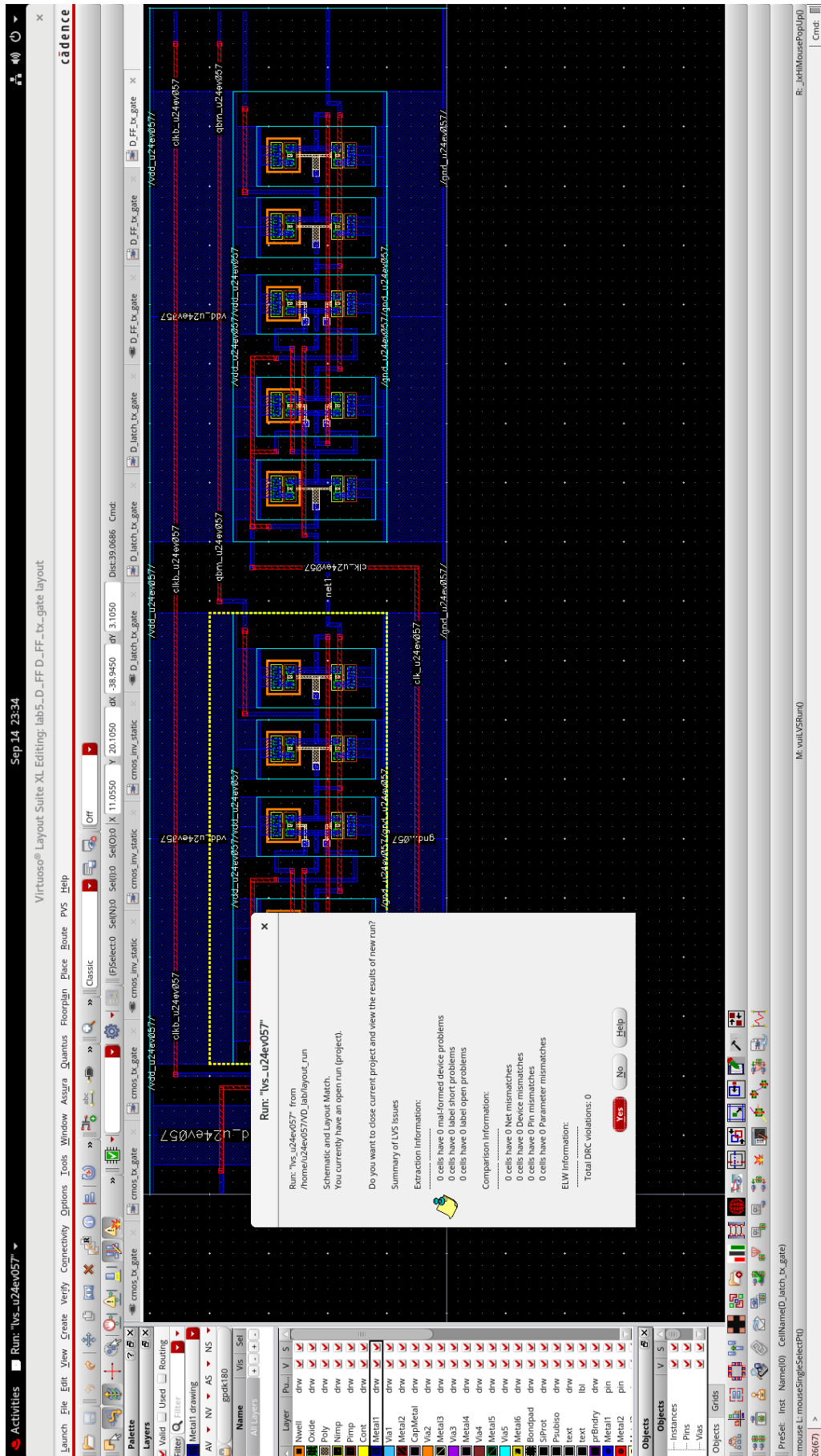


Figure .14: Tx Dff Lvs

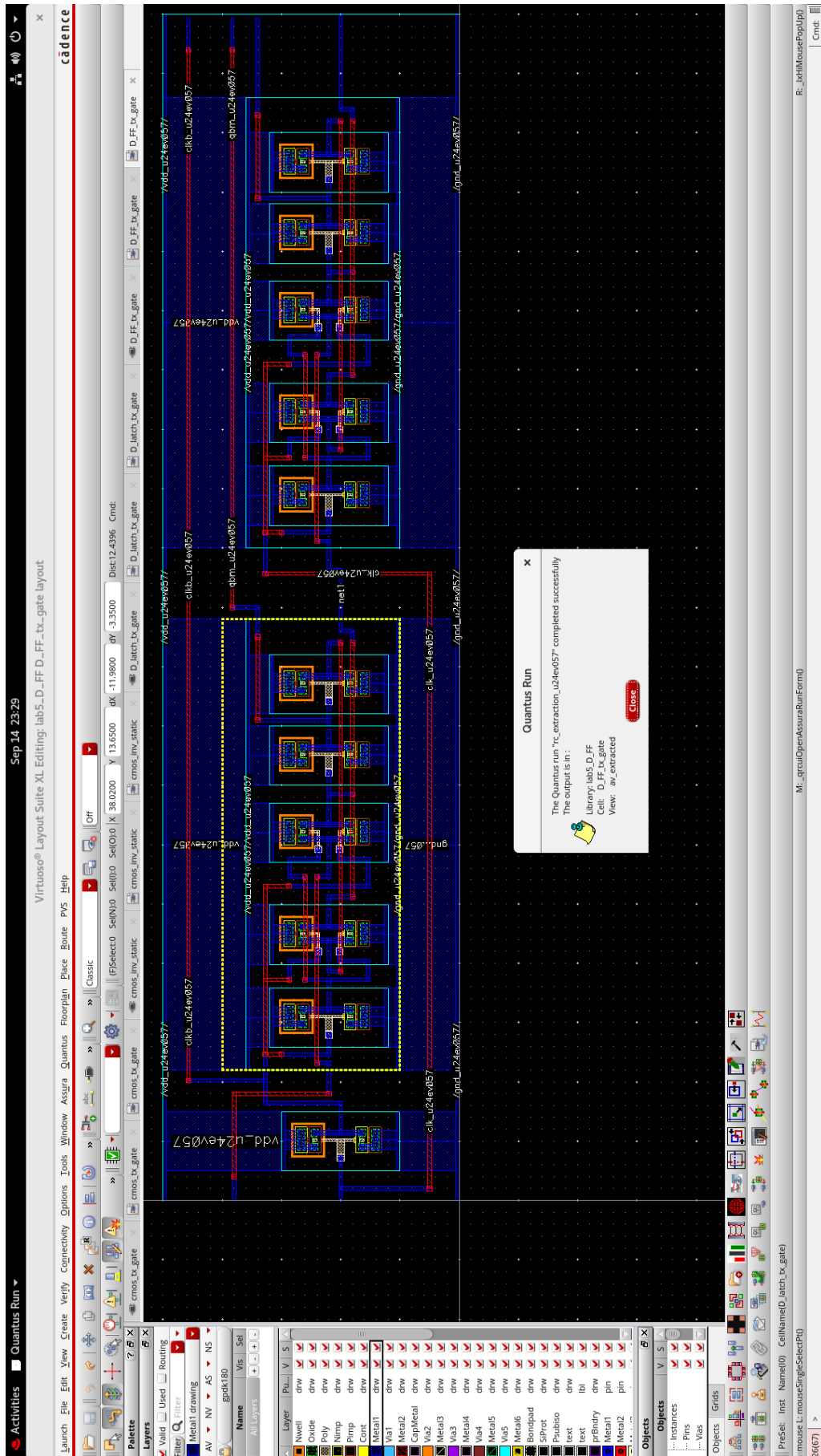


Figure .15: Tx Dff Quantus

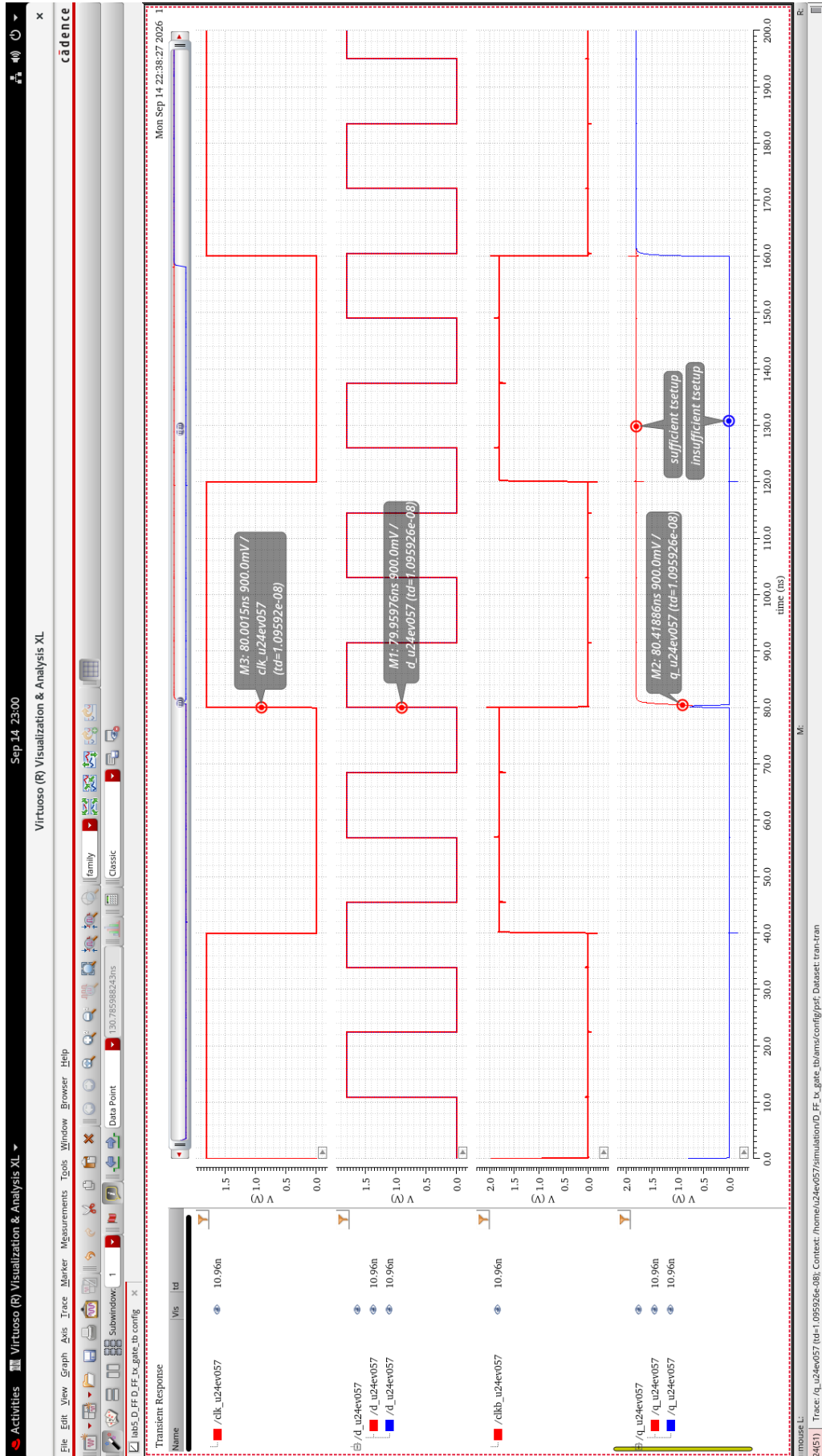


Figure .16: Tx Tran Pre Layout

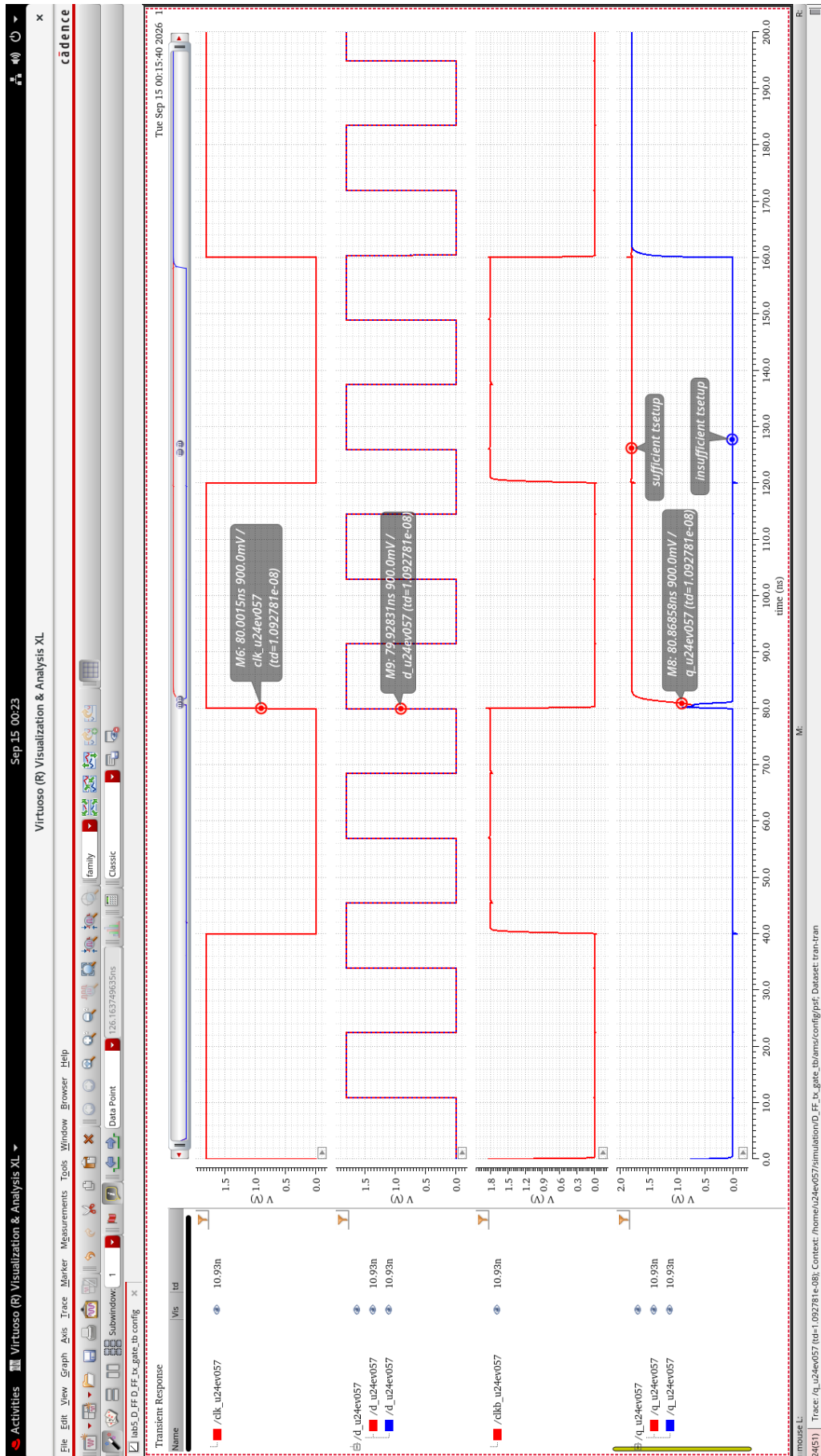


Figure .17: Tx Tran Post Layout

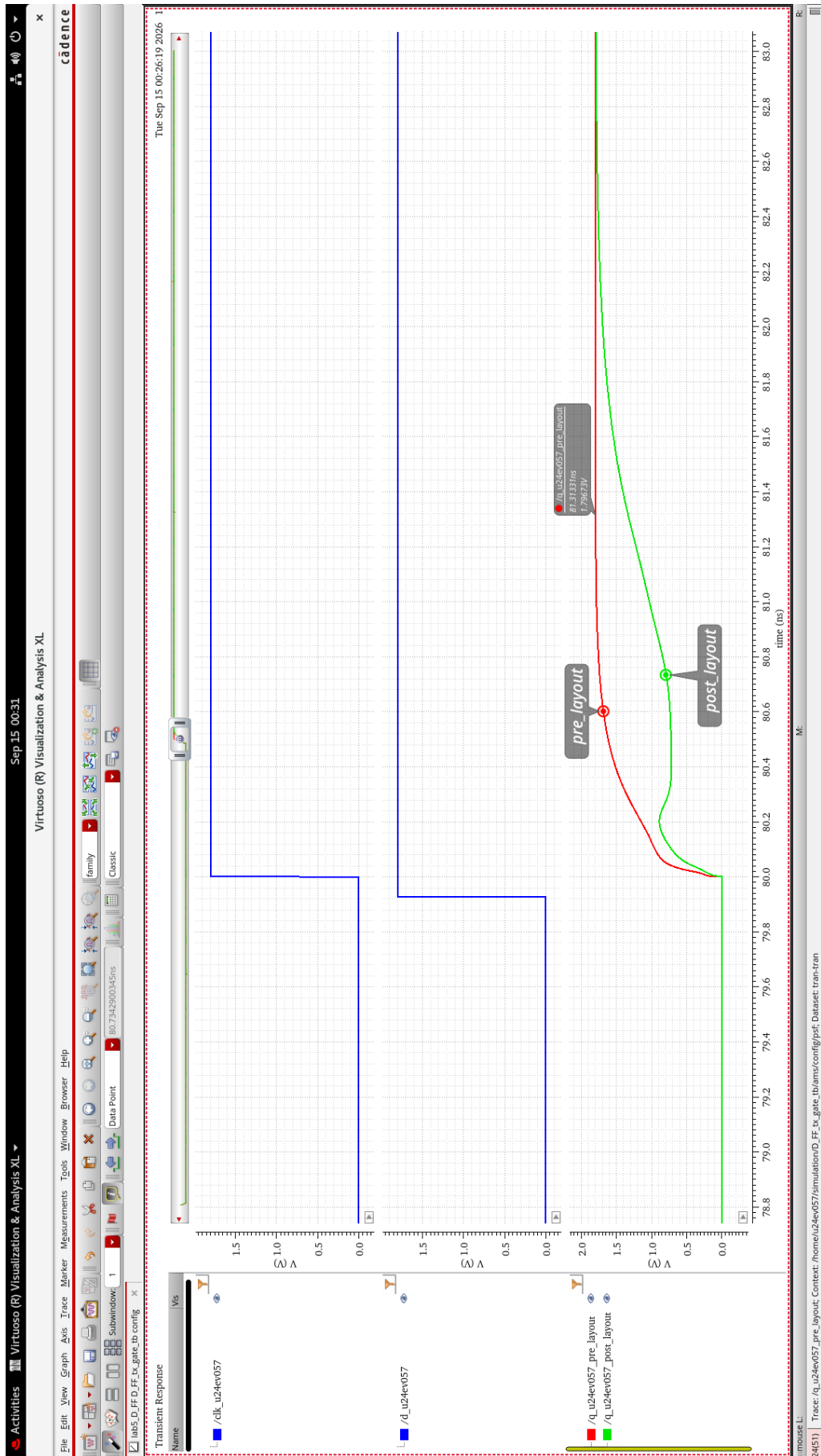


Figure .18: Tx Tran Pre Vs Post

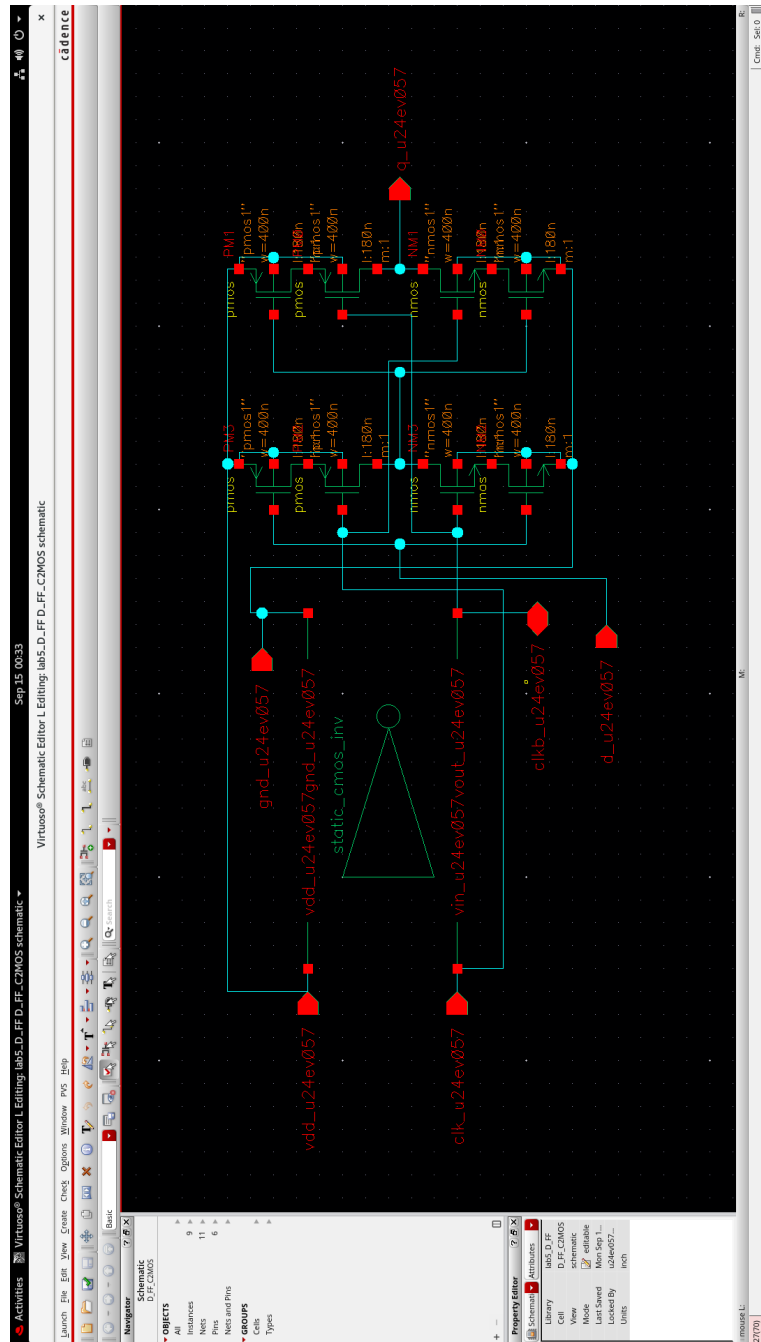


Figure .19: C2mos Dff Schematic

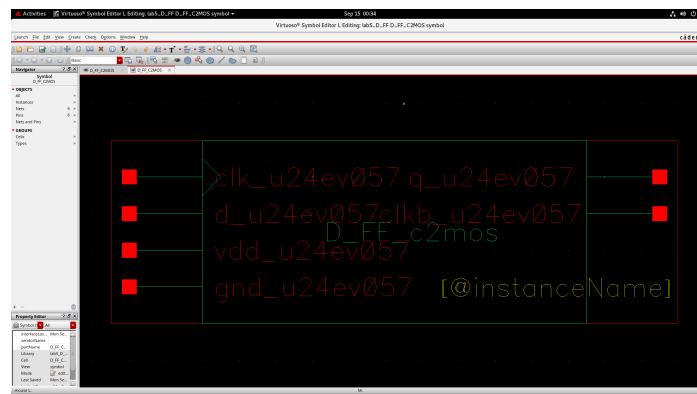


Figure .20: C2mos Dff Symbol

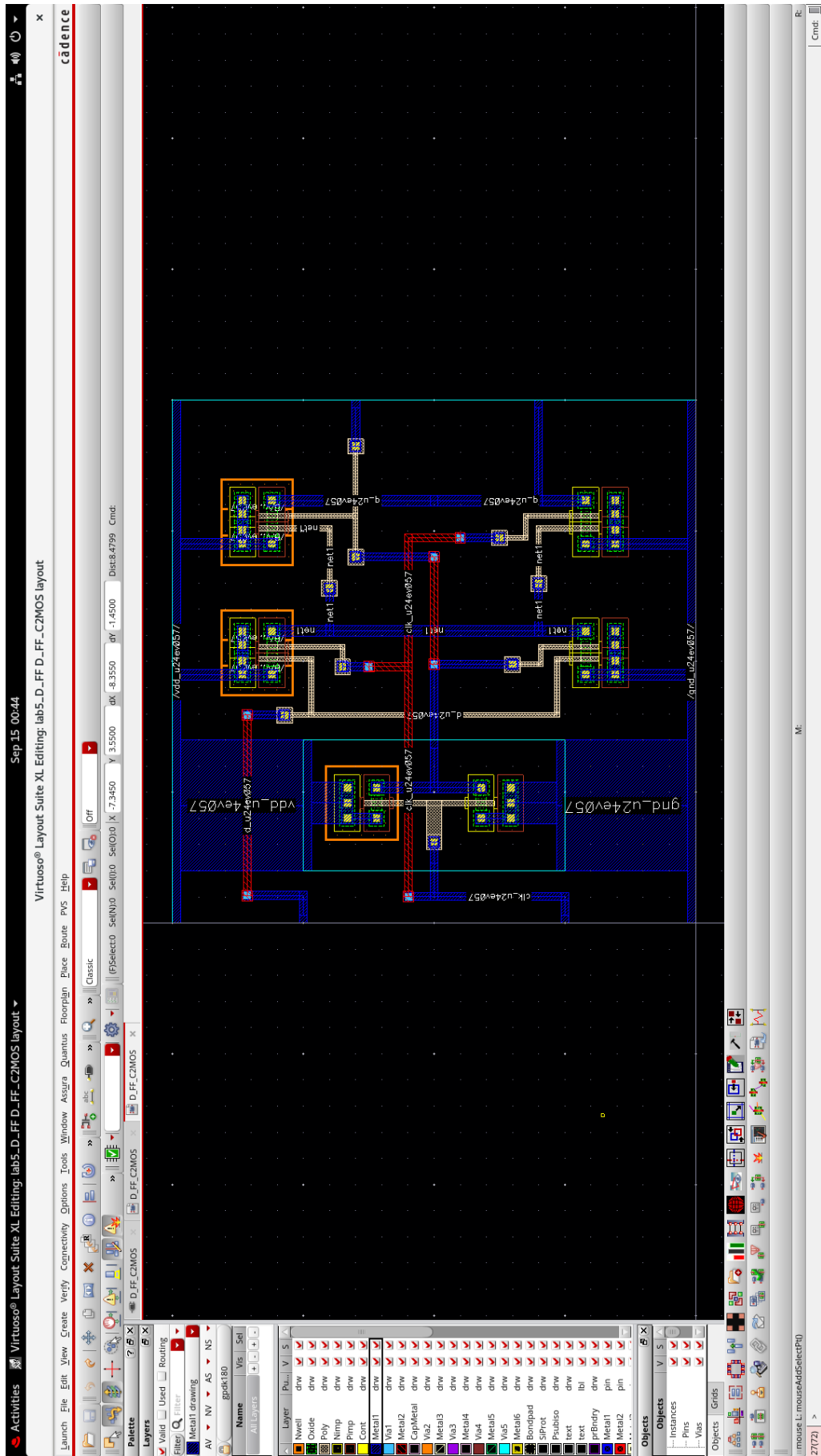


Figure .21: C2mos Dff Layout

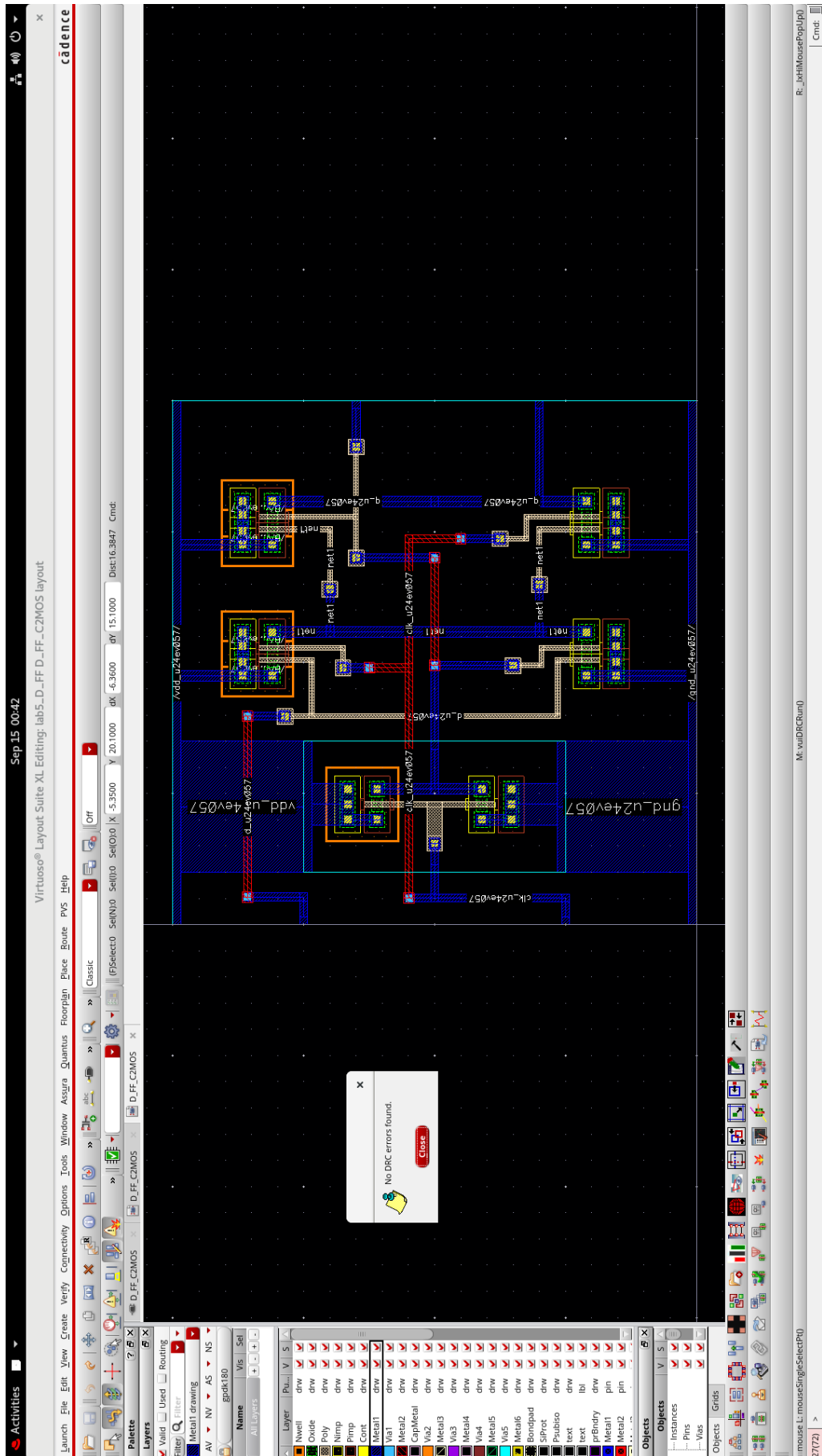


Figure .22: C2mos Dff Drc

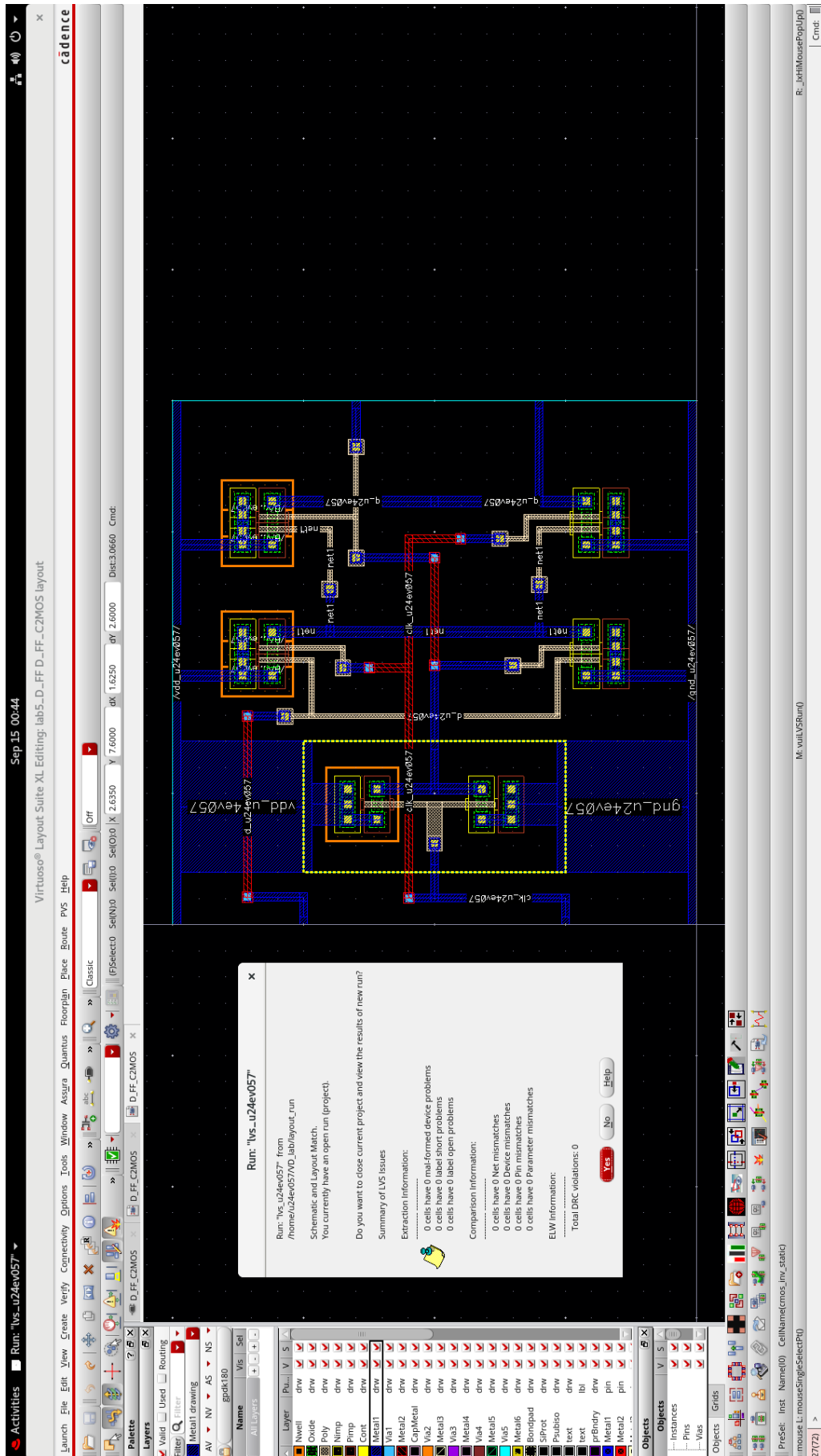


Figure .23: C2mos Dff Lvs

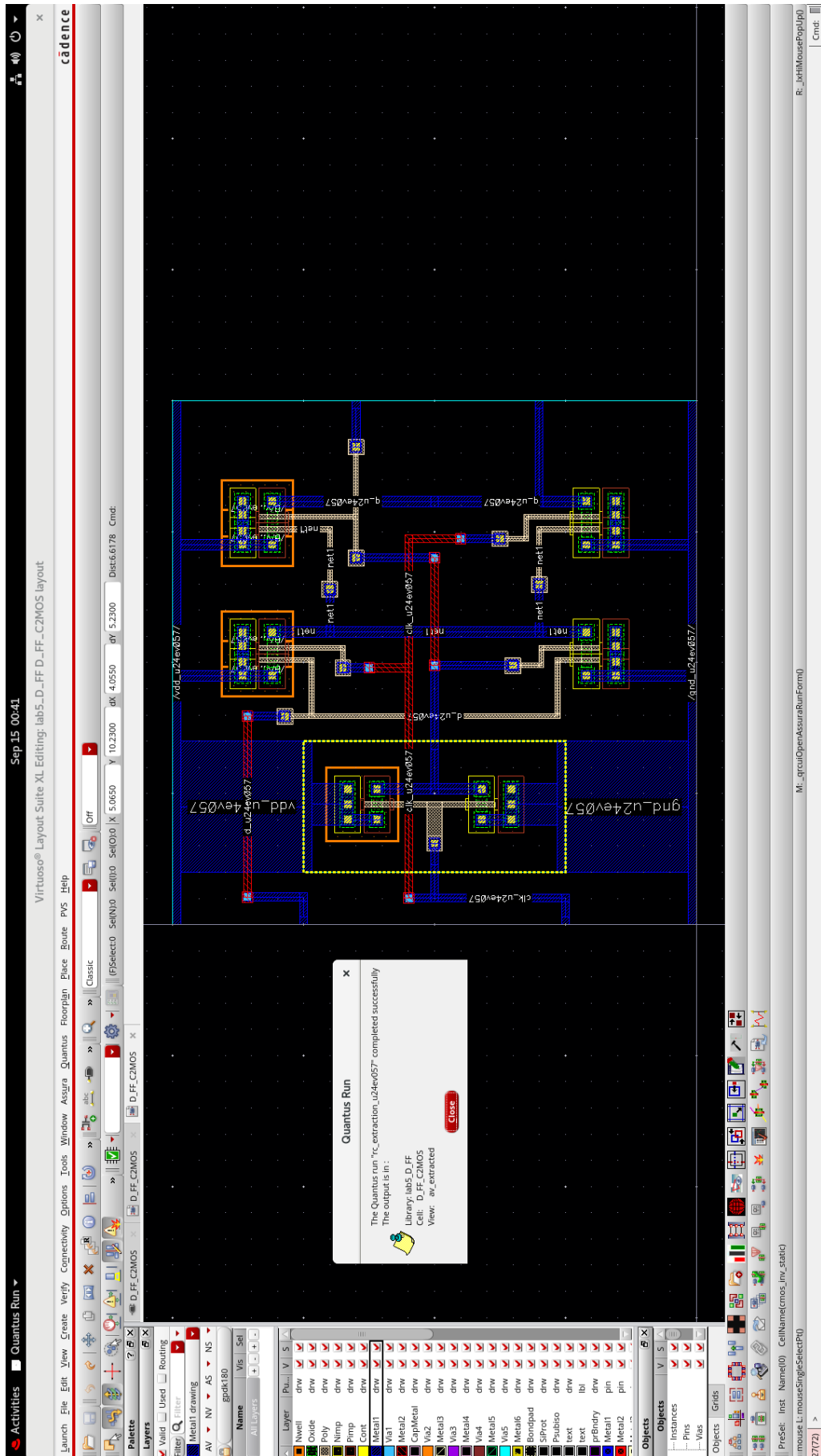


Figure .24: C2mos Dff Quantus

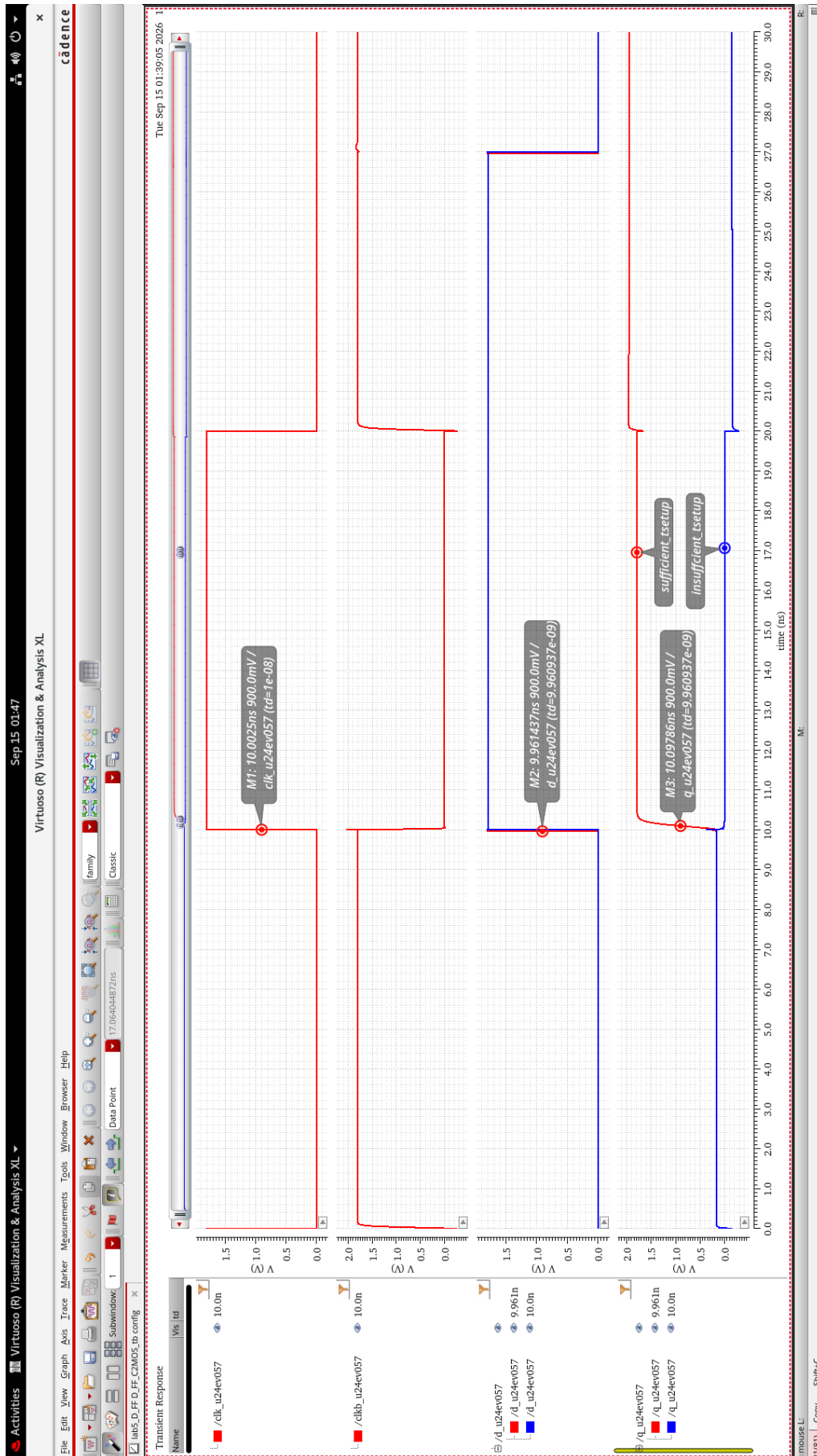


Figure .25: C2mos Dff Tran Pre Layout

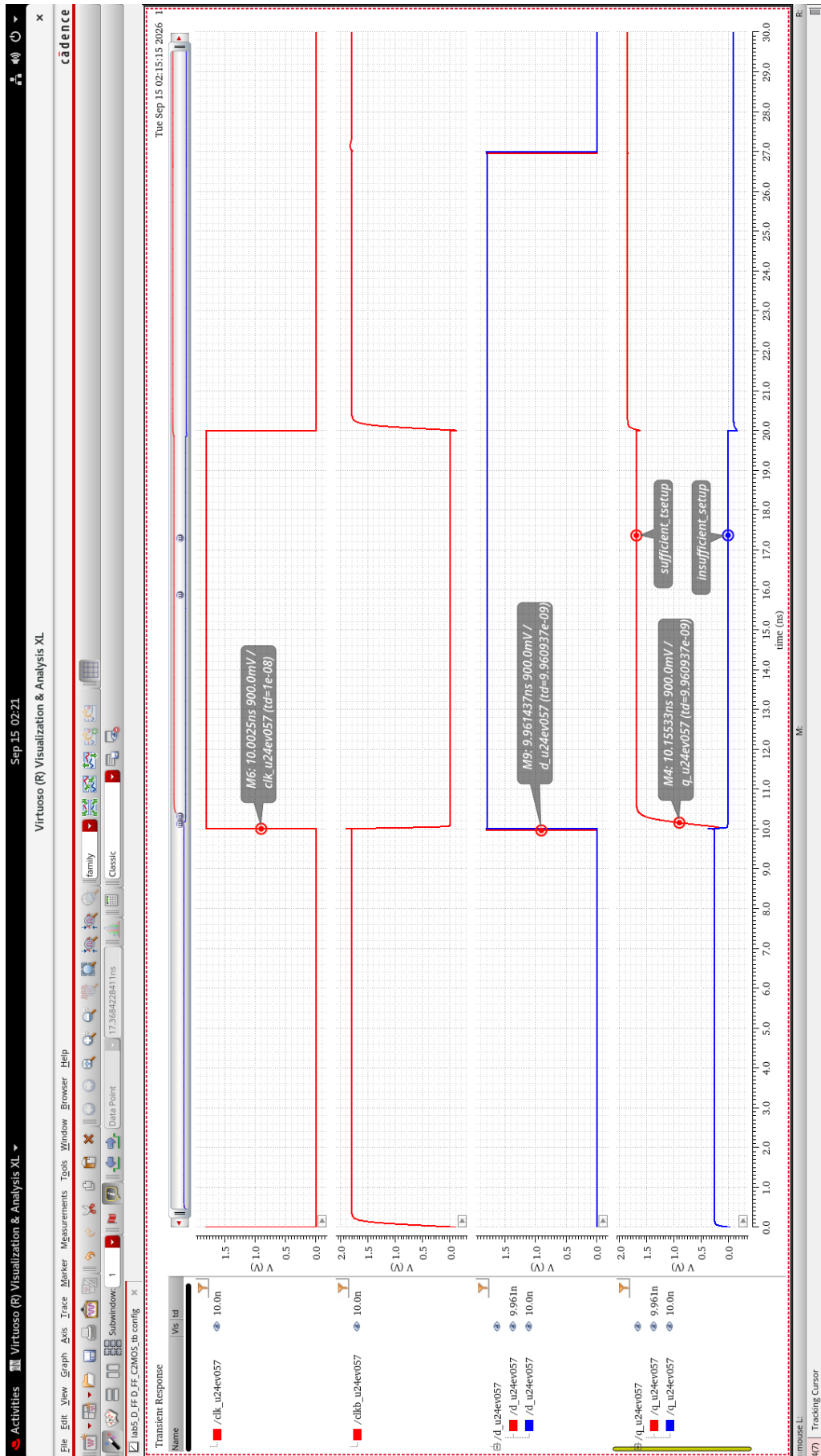


Figure .26: C2mos Tran Post Layout

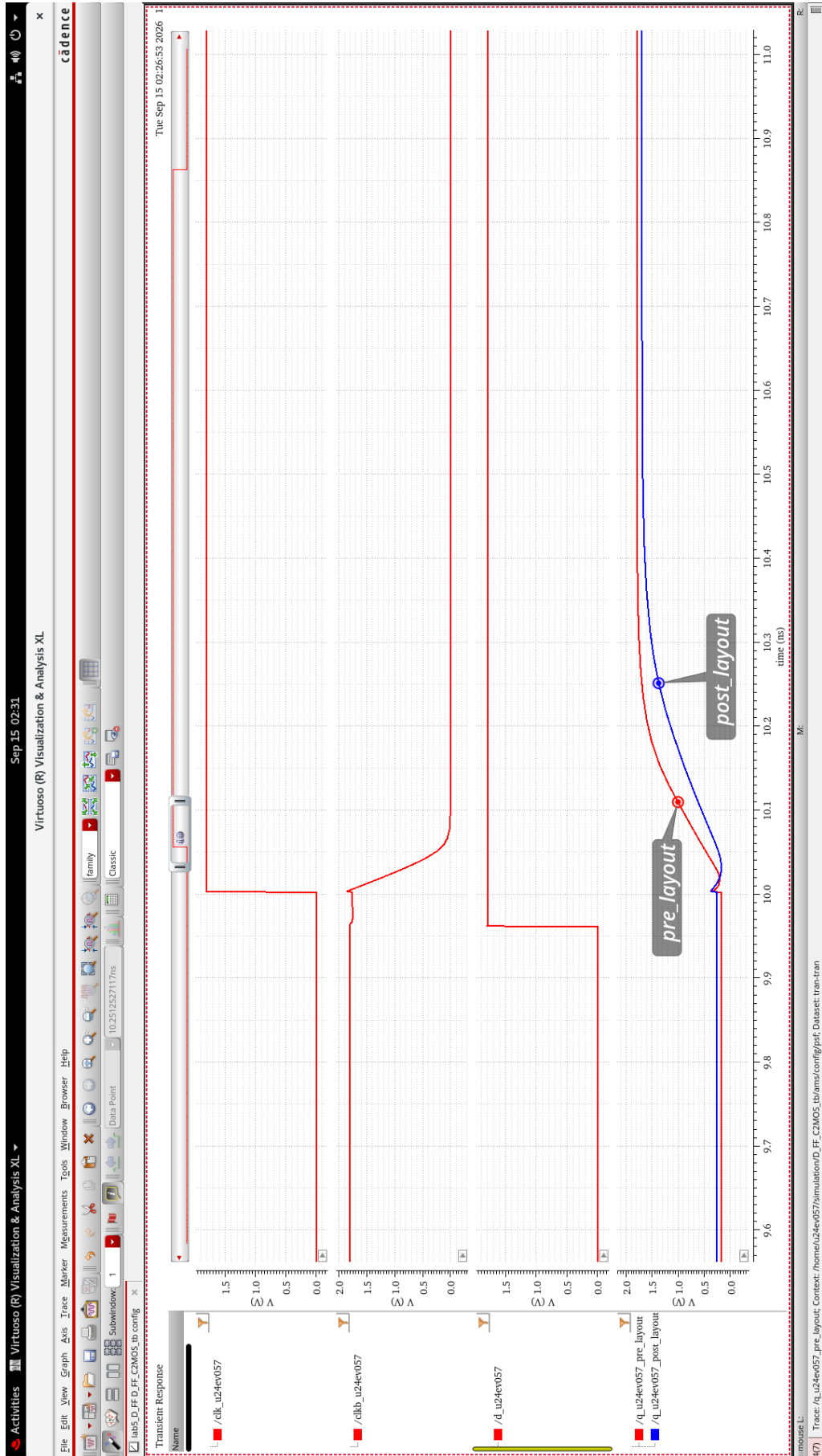


Figure .27: C2mos Tran Pre Vs Post

7. RESULT

The 2-input Dynamic CMOS NAND gate was successfully designed and verified using 180 nm CMOS technology, and the pre-layout and post-layout simulation results were compared to evaluate the effect of layout parasitic on circuit performance.